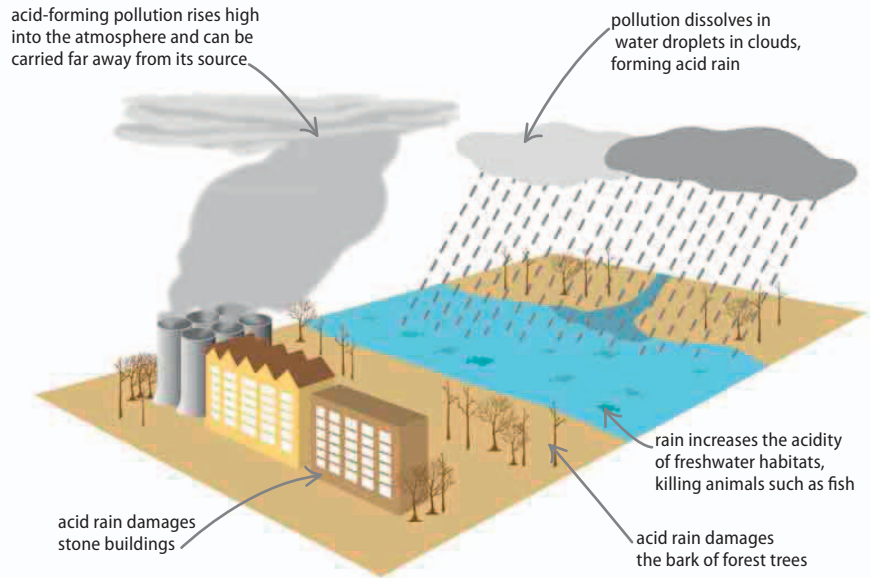


Acid rain

All rainwater is slightly acidic. This is because carbon dioxide in the air dissolves in it, making weak carbonic acid (as found in carbonated drinks). However, sometimes oxides of sulfur and nitrogen are released into the atmosphere as industrial waste. When these dissolve in water, they form much more potent acids, which have a damaging effect on wildlife when they fall as rain.

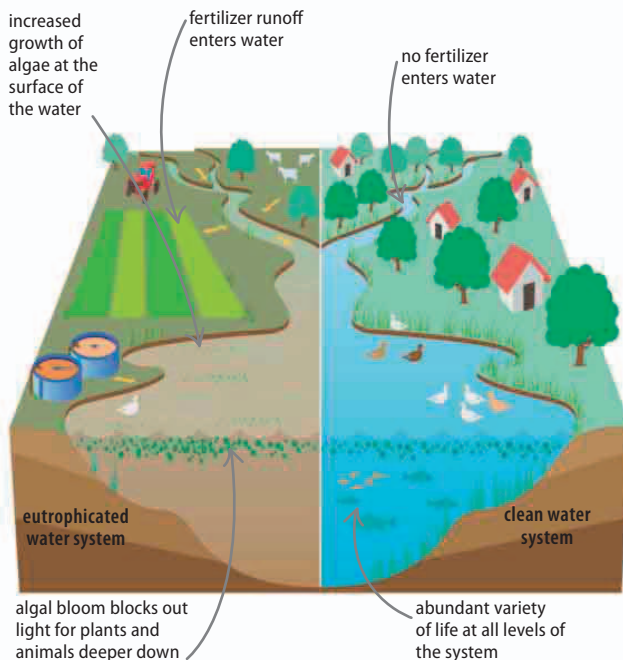
▷ Gases released

Coal-burning power plants and engines fueled by oil or gasoline release gases that can form acid rain. The rain often falls far from its source. It has many effects, including killing animal and plant life and damaging buildings.



Eutrophication

Fertilizers provide crops with nutrients such as nitrates and phosphates. These compounds boost the growth of any plant, and cause thick blooms of algae if they are washed by heavy rains into lakes and rivers. This leads to a process of eutrophication, where these blooms choke out life in the water.



Biomagnification

Even when present in tiny amounts, some pollutants can have an impact through a process called biomagnification. When an animal cannot break down a chemical, it is stored in its body, and is passed on to any predator that eats it. The concentration of this pollutant in animal tissues increases at each stage of the food chain, reaching damaging levels in top predators.

▷ DDT disaster

The biomagnification of an insecticide called DDT nearly wiped out many birds of prey in the USA in the 1940s and 1950s. DDT was thought harmless to vertebrates, but it built up in the bodies of fish and other animals in their environments. DDT poisoning resulted in birds, such as ospreys and bald eagles, laying eggs with very thin shells, so many eggs smashed in the nest.

